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OR Table

x_1	x_2	OR Output y
0	0	0
0	1	1
1	0	1
1	1	1

$w_1 = 0.2,\ w_2 = 0.1,\ w_0 = 0.1,\ \eta = 0.1$

for $(\overset{x_1, x_2}{0, 0})$ $y = 0$

$z = w_0 + x_1 w_1 + x_2 w_2$

$= 0.1 + 0 \times 0.2 + 0 \times -0.1$

$= 0.1$

$\hat{y} = 1\ [\because z \geq 0]$

$w_1 = w_1 + \eta(y - \hat{y})x_1$

$= 0.2 + 0.1(0-1) \times 0$

$= 0.2$

$w_2 = w_2 + \eta(y - \hat{y})x_2$

$= -0.1 + 0.1(0-1) \times 0$

$= -0.1$

$w_0 = w_0 + \eta(y - \hat{y})$

$= 0.1 + 0.1(0-1)$

$= 0.1 - 0.1$

$= 0$

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$(1,0)$ $y=1$ $\hat{y}=0$.

for $(0,1)$, $y=1$

$z = w_0 + x_1 w_1 + x_2 w_2$

$= 0.1 + 0 \times 0.2 + 1 \times -0.1$

$= 0.1 - 0.1$

$= 0$

$\hat{y} = 0$ $[\because z \geq 0]$

$w_1 = w_1 + \eta(y - \hat{y}) x_1$

$= 0.2 + 0.1(1-0) \times 0$

$= 0.2$

$w_2 = w_2 + \eta(y - \hat{y}) x_2$

$= -0.1 + 0.1(1-0) \times 1$

$= -0.1 + 0.1$

$= 0$

$w_0 = w_0 + \eta(y - \hat{y})$

$= 0 + 0.1(1-0)$

$= 0.1$

for $(1,0)$ $y=1$

$z = w_0 + x_1 w_1 + x_2 w_2$

$= 0.1 + 1 \times 0.2 + 0 \times 0$

$= 0.3$

$\hat{y} = 1$ $[\because z \geq 0]$

$w_1 = w_1 + \eta(y - \hat{y}) x_1$

$= 0.2 + 0.1(1-1) \times 1$

$= 0.2 + 0.1$

$= 0.1$

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$w_2 = w_2 + \eta(y-\hat{y})x_2$
$= 0 + 0.1(0-1)\times 1$
$= -0.1$

$w_0 = w_0 + \eta(y-\hat{y})$
$= 0.1 + 0.1(0-1)$
$= 0$

for (1,1) $y = 1$

$z = w_0 + x_1 w_1 + x_2 w_2$
$= 0.2 + 1\times 0.3 + 1\times 0.1$
$= 0.6$
$\hat{y} = 1 \; [\because z \geq 0]$
$w_1 = w_1 + \eta(y-\hat{y})x_1$
$= 0.3 + 0.1(1-0)$

$0 + 1\times 0.1 + 1\times -0.1$
$0.1 - 0.1$
0
$\hat{y} = 0 \; [\because z \geq 0]$

$w_1 = w_1 + \eta(y-\hat{y})x_1$
$= 0.1 + 0.1(1-0)\times 1$
$= 0.2$

$w_2 = w_2 + \eta(y-\hat{y})x_2$
$= -0.1 + 0.1(1-0)\times 1$
$= 0$

$w_0 = w_0 + \eta(y-\hat{y})$
$= 0 + 0.1(1-0)$
$= 0.1$

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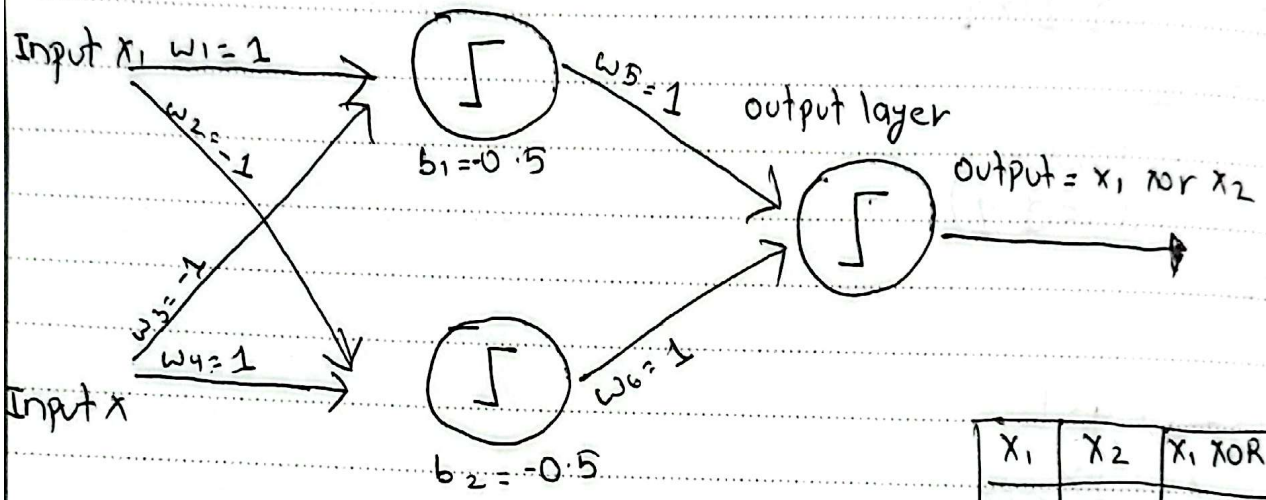
Solving "XOR" with multilayer Perception



x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0

$x_1 = 0, x_2 = 0$

$z_1 = b_1 + x_1 w_1 + x_2 w_3$

$= -0.5 + 0 \times 1 + 0 \times -1$

$= -0.5$

$h_{1out} = 0 \; [\because z_1 < 0]$

$z_2 = b_2 + x_1 w_2 + x_2 w_4$

$= -0.5 + 0 \times -1 + 0 \times 1$

$= -0.5$

$h_{2out} = 0$

$z_3 = h_1 w_5 + h_2 w_6 + b_3$

$= 0 \times 1 + 0 \times 1 + (-0.5)$

$= -0.5$

$O_{out} = 0$

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$x_1 = 0, x_2 = 1$

$z_1 = b_1 + x_1 w_1 + x_2 w_3$

$= -0.5 + 0 \times 1 + 1 \times -1$

$= -0.5 - 1$

$= -1.5$

$h_{1out} = 0 \; [\because z < 0]$

$z_2 = b_2 + x_1 w_2 + x_2 w_4$

$= -0.5 + 0 + 1 \times 1$

$= 0.5$

$h_{2out} = 1$

$z_3 = h_1 w_5 + h_2 w_6 + b_3$

$= 0 \times 1 + 1 \times 1 + (-0.5)$

$= 1 - 0.5$

$O_{out} = 1$

$x_1 = 1, x_2 = 0$

$z_1 = b_1 + x_1 w_1 + x_2 w_3$

$= -0.5 + 1 \times 1 + 1 \times 0$

$= 0.5$

$h_{out} = 1 \; [\because z < 0]$

$z_2 = b_2 + x_1 w_2 + x_2 w_4$

$= -0.5 + 1 \times -1 + 0 \times 1$

$= -1.5$

$h_{2out} = 0$

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$z_3 = h_1 w_5 + h_2 w_6 + b_3$

$= 1 \times 1 + 0 \times 1 + (-0.5)$

$= 1 - 0.5$

$= 0.5$

O_{out} 1

$x_1 = 1, x_2 = 1$

$z_1 = b_1 + x_1 w_1 + x_2 w_2$

$= -0.5 + 1 \times 1 + 1 \times -1$

$= -0.5$

$h_{1out} = 0$

$z_2 = b_2 + x_1 w_2 + x_2 w_4$

$= -0.5 + 1 \times -1 + 1 \times 1$

$= -0.5$

$h_{2out} = 0$

$z_3 = h_1 w_5 + h_2 w_6 + b_3$

$= 0 \times 1 + 0 \times 1 + (-0.5)$

$= -0.5$

$h_{3out} = 0$